

Question #	1	2	3	4	5	6	7	8	9	10	Total
Assigned Marks											
Obtained Marks											

Final paper Rubric

$$x_0 = 0.5$$

$$x_1 = 1$$

$$x_2 = 2$$

$$y_0 = 4$$

$$y_1 = 3$$

$$y_2 = 2$$

As these data points are unequally spaced, let us use Lagrange interpolation to generate a polynomial $P_2(x)$ interpolating them.

$$P_2(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} y_2$$

$$= \frac{(x-1)(x-2)}{(0.5-1)(0.5-2)} (4) + \frac{(x-0.5)(x-2)}{(1-0.5)(1-2)} (3) + \frac{(x-0.5)(x-1)}{(2-0.5)(2-1)} (2)$$

$$= \frac{(x^2-3x+2)}{(-0.5)(-1.5)} (4) + \frac{(x^2-2.5x+1)}{(0.5)(-1)} (3) + \frac{(x^2-1.5x+0.5)}{(1.5)(1)} (2)$$

$$= \frac{y}{0.75} (x^2 - 3x + 2) - 3 \frac{(x^2 - 2.5x + 1)}{0.5} + 2 \frac{(x^2 - 1.5x + 0.5)}{1.5}$$

$$= \frac{x^2 y}{0.75} - \frac{3xy}{0.75} + \frac{2y}{0.75} - 3x^2 + \frac{7.5x}{0.5} - 3 + \frac{2x^2}{1.5} - \frac{3x}{1.5} + \frac{1}{1.5}$$

$$= \frac{4}{3} x^2 y - 4xy + \frac{8y}{3} - 6x^2 + 15x - 6 + \frac{4x^2}{3} - 2x + \frac{2}{3}$$

$$= \left(\frac{4}{3} y - 6 + \frac{4}{3} \right) x^2 + (-4y + 15 - 2)x + \left(\frac{8y}{3} - 6 + \frac{2}{3} \right)$$

We know that the coefficient of x^2 is 6, so

$$\frac{4}{3} y - 6 + \frac{4}{3} = 6 \Rightarrow \frac{4y}{3} = 6 + 6 - \frac{4}{3} = \frac{32}{3}$$

$$y = \frac{32}{3} \times \frac{3}{4}$$

$$\boxed{y = 8}$$

If $y = 8$, then the polynomial becomes:

$$P_2(x) = \left[\frac{4(8)}{3} - 6 + \frac{4}{3} \right] x^2 + (-4(8) + 15 - 2)x +$$

$$\left[\frac{8(8)}{3} - 6 + \frac{2}{3} \right]$$

$$\Rightarrow P_2(x) = 6x^2 - 19x + 16$$

Verification : $P_2(0.5) = 8$; $P_2(1) = 3$; $P_2(2) = 2$.

Q2 :

$$\theta(t) = L \frac{di}{dt} + Ri, \quad L = 0.98, R = 0.142$$

let $x = t$ & $y = i$

Using Newton Backward difference formula.

$$y_p = y_n + P \nabla y_n + \frac{P(P+1)}{2!} \nabla^2 y_n + \frac{P(P+1)(P+2)}{3!} \nabla^3 y_n + \dots$$

where $P = \frac{x - x_n}{h}$

$$P = \frac{1.03 - 1.03}{0.01} = 0$$

$$P = 0$$

$$\frac{dy}{dx} = \frac{dy}{dP} \cdot \frac{dP}{dx}$$

①

$$\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_n + \frac{2P+1}{2} \nabla^2 y_n + \frac{3P^2+6P+2}{6} \nabla^3 y_n + \dots \right]$$

Now, in the given situation, we need $\frac{di}{dt}$ at $t = 1.03$ i.e. $\left. \frac{dy}{dx} \right|_{x=1.03}$

The data points given to us is:

$x = t$	$y = i$			
$x_0 = 1.00$	3.10	0.02	0	0.02
$x_1 = 1.01$	3.12	0.02	0.02	
$x_2 = 1.02$	3.14	0.04		
$x_3 = 1.03$	3.18			

① $\Rightarrow$

$$\frac{di}{dt} = \frac{1}{0.01} \left[\frac{0.04 + 2(0) + 1}{2} (0.02) + \frac{3(0)^2 + 6(0) + 2}{6} (0.02) \right]$$

$$= \frac{1}{0.01} \times \frac{17}{300}$$

$$\boxed{\frac{di}{dt} = \frac{17}{3}}$$

~~As you~~ Now, $v(t) = L \frac{di}{dt} + Ri$

$$\Rightarrow v(1.03) = 0.98 \left(\frac{17}{3} \right) + (0.142)(3.18)$$

$$\boxed{v(1.03) = 6.005}$$

Q3:

$$\text{Let } y(x) = 2\pi g(x) \sqrt{1 + [g'(x)]^2}$$

$$\text{As } g(x) = e^{-x}$$

$$g'(x) = -e^{-x}$$

$$[g'(x)]^2 = e^{-2x}$$

Thus we have

$$y(x) = 2\pi e^{-x} \sqrt{1 + e^{-2x}}$$

Using steps of size 0.2 from 0 to 1, we get.

$x_0 = 0$	$y_0 = 8.885766$
$x_1 = 0.2$	$y_1 = 6.648456$
$x_2 = 0.4$	$y_2 = 5.070439$
$x_3 = 0.6$	$y_3 = 3.933455$
$x_4 = 0.8$	$y_4 = 3.095122$
$x_5 = 1$	$y_5 = 2.462904$

Trapezoidal Rule

$$\int_a^b y(x) dx = \frac{b-a}{2} [y_0 + y_5 + 2(y_1 + y_2 + y_3 + y_4)]$$

$$= 0.1 [13.348670 + 2(18.747472)]$$

Surface Area. = 4.8843614

$$x = \sqrt[3]{N}, \quad x^3 = N \Rightarrow x^3 - N = 0$$

Q4: let $f(x) = x^3 - N$
then

$$f'(x) = 3x^2$$

let us using the Newton Raphson method

$$x_{n+1} = x_n - \frac{f(x)}{f'(x)}$$

$$= x_n - \frac{x_n^3 - N}{3x_n^2}$$

$$= x_n - \frac{x_n^3}{3x_n^2} + \frac{N}{3x_n^2}$$

$$= \frac{3x_n^3 - x_n^3}{3x_n^2} + \frac{N}{3x_n^2}$$

$$= \frac{2x_n}{3} + \frac{N}{3x_n^2}$$

$$x_{n+1} = \frac{1}{3} \left(\frac{2x_n + \frac{N}{x_n^2}}{1} \right) \quad \text{--- (2)}$$

Now, finding cube root of 15 correct to three decimal places

$$\text{So } N = 15$$

$$\text{As } f(x) = x^3 - N$$

$$\Rightarrow f(x) = x^3 - 15$$

$$\text{let } a = 2 \text{ as } f(a) < 0$$

$$\& b = 3 \text{ as } f(b) > 0$$

$$\text{let } x_0 = 2.5$$

Using (2)

$$x_{n+1} = \frac{1}{3} \left[\frac{2x_n + \frac{15}{x_n^2}}{1} \right]$$

$$= \frac{1}{3} \left[\frac{2x_0 + \frac{15}{x_0^2}}{1} \right]$$

$$x_1 = 2.467$$

$$x_2 = 2.466$$

$$x_3 = 2.466$$

So, cube root of 15 correct to 3-decimal places is 2.466.

$$\text{Verification : } (2.466)^3 = 14.996 \approx 15$$

Q5:

We have

$$AX = B \text{ as}$$

$$\begin{bmatrix} 4 & 1 & -1 \\ -1 & 3 & 1 \\ 2 & 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ 1 \end{bmatrix}$$

We use Doolittle's method (as given in q5)
i.e consider $\delta_{ii} = 1$.

$$\begin{bmatrix} 4 & 1 & -1 \\ -1 & 3 & 1 \\ 2 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \delta_{21} & 1 & 0 \\ \delta_{31} & \delta_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ \delta_{21}u_{11} & \delta_{21}u_{12} + u_{22} & \delta_{21}u_{13} + u_{23} \\ \delta_{31}u_{11} & \delta_{31}u_{12} + \delta_{32}u_{22} & \delta_{31}u_{13} + \delta_{32}u_{23} + u_{33} \end{bmatrix}$$

$$u_{11} = 4 \quad \delta_{21} = \frac{-1}{u_{11}} = -\frac{1}{4}$$

$$u_{12} = 1$$

$$u_{13} = -1 \quad \delta_{31} = \frac{2}{u_{11}} = \frac{1}{2}$$

$$u_{22} = 3 - \delta_{21}u_{12} = \frac{13}{4}$$

$$\delta_{32} = \frac{2 - \delta_{31}u_{12}}{u_{22}} = \frac{3 \times 4}{2 \times 13} = \frac{6}{13}$$

$$u_{23} = 1 - \delta_{21}u_{13} = \frac{3}{4}$$

$$u_{33} = 5 - \delta_{31}u_{13} - \delta_{32}u_{23}$$

$$u_{33} = 5 + \frac{1}{2} - \frac{9}{26} = \frac{67}{13}$$

So, we have

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & 0 \\ 0.5 & 6/13 & 1 \end{bmatrix}, U = \begin{bmatrix} 4 & 1 & -1 \\ 0 & 13/4 & 0.75 \\ 0 & 0 & 67/13 \end{bmatrix}$$

Now, $AX = B$
 $\Rightarrow LUX = B$

Let, $UX = Z$, then $LZ = B$

We find Z from $LZ = B$ then use it to find x from $UX = Z$

$$\begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & 0 \\ 0.5 & 6/13 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ 1 \end{bmatrix}$$

Using forward substitution

$$z_1 = 5$$

$$z_2 = -11/4$$

$$z_3 = -3/13$$

Now, we have

$$\begin{bmatrix} 4 & 1 & -1 \\ 0 & 13/4 & 0.75 \\ 0 & 0 & 67/13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -11/4 \\ 3/13 \end{bmatrix}$$

Using back-substitution, we get

$$x_1 = 1.4478$$

$$x_2 = -0.8358$$

$$x_3 = -0.04478$$

So,

$$x = 1.45$$

$$y = -0.84$$

$$z = -0.045$$

Q6:

let $x = t$

$$\text{We have } \frac{dy}{dx} = 1 - 0.5y - 0.1yx$$

$$y(0) = 5$$

Using Picard Method.

$$y_1^{(1)} = y_0 + \int_{x_0}^x f(x, y_0) dx$$

$$= 5 + \int_0^x (1 - 0.5y_0 - 0.1y_0x) dx$$

$$= 5 + \int_0^x [1 - 0.5(5) - 0.1(5)x] dx$$

$$= 5 + \int_0^x (-1.5 - 0.5x) dx$$

$$= 5 - [1.5x + 0.25x^2]_0^x$$

$$\boxed{y_1^{(1)} = 5 - 1.5x - 0.25x^2}$$

$$y_1^{(2)} = y_0 + \int_{x_0}^x f(x, y_1^{(1)}) dx$$

$$= 5 + \int_0^x [1 - 0.5(5 - 1.5x - 0.25x^2) - 0.1x(5 - 1.5x - 0.25x^2)] dx$$

$$= 5 + \int_0^x (1 - 2.5 + 0.75x + 0.125x^2 - 0.5x + 0.15x^2 + 0.025x^3) dx$$

$$= 5 + \int_0^x [-1.5 + 0.25x + 0.275x^2 + 0.025x^3] dx$$

$$= 5 + \left[-1.5x + \frac{0.125x^2}{2} + \frac{0.275x^3}{3} + \frac{0.025x^4}{4} \right]_0^x$$

$$y_1^{(2)} = 5 - 1.5x + 0.125x^2 + \frac{0.275x^3}{3} + \frac{0.025x^4}{4}$$

$$y_1^{(3)} = y_0 + \int_{x_0}^x f(x, y_1^{(2)}) dx$$

$$= 5 + \int_0^x [1 - 0.5y_1^{(2)} - 0.1y_1^{(2)}] dx$$

$$= 5 + \int_0^x \left\{ 1 - \left[2.5 - 0.75x + \frac{0.125x^2}{2} + \frac{0.275x^3}{6} + \frac{0.025x^4}{8} \right] - \left[0.5x - 0.15x^2 + \frac{0.0125x^3}{3} + \frac{0.0275x^4}{4} + \frac{0.0025x^5}{5} \right] \right\} dx$$

$$= 5 + \int_0^x \left[-1.5 + 0.25x + 0.0875x^2 - \frac{7x^3}{120} - \frac{59x^4}{4800} - \frac{0.0025x^5}{4} \right] dx$$

$$= 5 + \left[-1.5x + \frac{0.125x^2}{2} + \frac{0.0875x^3}{3} - \frac{7x^4}{480} - \frac{59x^5}{24000} - \frac{0.0025x^6}{24} \right]$$

$$y_1^{(3)} = 5 - 1.5x + 0.125x^2 + \frac{0.0875x^3}{3} - \frac{7x^4}{480} - \frac{59x^5}{24000} - \frac{0.0025x^6}{24}$$

For Residual,

$$\frac{dy}{dx} = 0 - 1.5 + 0.25x + 0.0875x^2 - \frac{7}{120}x^3 - \frac{59}{4800}x^4 - \frac{0.0025}{4}x^5$$

$$R(x) = \frac{dy}{dx} - f(x, y)$$

$$R(x) = -1.5 + 0.25x + 0.0875x^2 - \frac{7}{120}x^3 - \frac{59}{4800}x^4 - \frac{0.0025}{4}x^5 - 1 + 0.5y + 0.1yx \quad (*)$$

After putting the value of y in $(*)$, We get

$$y_1^{(3)}(0.25) = 4.633$$

$$y_1^{(3)}(0.5) = 4.2839$$

$$y_1^{(3)}(0.75) = 3.952$$

$$R(0.25) = 1.00096$$

$$R(0.5) = 1.0084$$

$$R(0.75) = 1.0304$$

Q7:

x	0	0.25	0.5	0.75	1
y	5	4.6332	4.28391	3.9524	?

Using $n=3$

$$y_{4,p} = y_0 + \frac{4(0.25)}{3} (2y'_1 - y'_2 + 2y'_3)$$

$$y'_1 = 1 - 0.5(4.6332) - 0.1(4.6332)(0.25)$$

$$y'_1 = -1.43243$$

$$y'_2 = 1 - 0.5(4.28391) - 0.1(4.28391)(0.5)$$

$$y'_2 = -1.35615$$

$$y'_3 = 1 - 0.5(3.9524) - 0.1(3.9524)(0.75)$$

$$y'_3 = -1.27263$$

$$y_{4,p} = 5 + \frac{1}{3} [2(-1.43243) - (-1.35615) + 2(-1.27263)]$$

$$y_{4,p} = 3.64868$$

$$y'_4 = 1 - 0.5(3.64868) - 0.1(3.64868)(1)$$

$$y'_4 = -1.189208$$

Corrector formula:

$$y_{4,c} = y_2 + \frac{h}{3} (y_2' + 4y_3' + y_4')$$
$$= 4.28391 + \frac{0.25}{3} (-1.35615 + 4(-1.27263) + (-1.189208))$$

$$y_{4,c} = 3.647587$$

Q8: $y'' - 2y' + 2y = e^{2x} \sin x$ — (1)
 $y(0) = -0.4, y'(0) = -0.6$

Sol let $\frac{dy}{dx} = z, \frac{d^2y}{dx^2} = \frac{dz}{dx}$

$$(1) \Rightarrow z' = 2z - 2y + e^{2x} \sin x.$$

So, we have

$$\begin{cases} \frac{dy}{dx} = f(x, y, z) = z \\ \frac{dz}{dx} = g(x, y, z) = 2z - 2y + e^{2x} \sin x \end{cases}$$
$$y(0) = -0.4, z(0) = -0.6$$

$$k_1 = h f(x_0, y_0, z_0)$$

$$= (0.2)(-0.6)$$

$$k_1 = -0.12$$

$$l_1 = h g(x_0, y_0, z_0)$$

$$= (0.2) [2z_0 - 2y_0 + e^{2x_0} \sin x_0]$$

$$= (0.2) [2(-0.6) - 2(-0.4) + e^{2(0)} \sin(0)]$$

$$l_1 = -0.08$$

$$k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right)$$

$$= (0.2) \left(-0.6 + \frac{(-0.08)}{2}\right)$$

$$k_2 = -0.128$$

$$l_2 = h g\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right)$$

$$= (0.2) g(0.1, -0.46, -0.64)$$

$$l_2 = -0.07157$$

$$k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right)$$

$$= 0.2 \left(z_0 + \frac{l_2}{2}\right)$$

$$= 0.2 \left(-0.6 - \frac{0.07157}{2}\right)$$

$$k_3 = -0.127157$$

$$l_3 = h g \left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2} \right)$$

$$= (0.2) g (0.1, -0.464, -0.635785)$$

$$= (0.2) \left[2(-0.635785) - 2(-0.464) + e^{2(-0.1)} \sin(0.1) \right]$$

$$= (0.2) (-0.341438)$$

$$l_3 = -0.06829$$

$$k_4 = h f(x_0 + h, y_0 + k_3, z_0 + l_3)$$

$$= (0.2) [-0.6 + (-0.06829)]$$

$$k_4 = -0.133658$$

$$l_4 = h g(x_0 + h, y_0 + k_3, z_0 + l_3)$$

$$= (0.2) g(0.2, -0.527157, -0.66829)$$

$$l_4 = -0.05541$$

$$\Delta y = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$= \frac{1}{6} [-0.12 + 2(-0.128) + 2(-0.127157) - 0.133658]$$

$$\Delta y = -0.12733$$

$$y_1 = y_0 + \Delta y$$

$$= -0.4 - 0.12733$$

$$y_1 = -0.5273$$

Q9:

As we know that

$$y'_i = \frac{y_{i+1} - y_{i-1}}{2h}$$

$$y''_i = \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}$$

$$y'' + 2y' + y = 0; \quad y(0) = 0; \quad y(1) = 1$$

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + 2 \frac{y_{i+1} - y_{i-1}}{2h} + y_i = 0 \quad \text{where } i=1, 2, 3, 4$$

$i=1$

$$\frac{1}{0.2^2} (y_2 - 2y_1 + y_0) + \frac{y_2 - y_0}{0.2} + y_1 = 0$$

$$25y_2 - 50y_1 + 25y_0 + 5y_2 - 5y_0 + y_1 = 0$$

$$30y_2 - 49y_1 + 20y_0 = 0$$

$i=2$

$$25(y_3 - 2y_2 + y_1) + 5(y_3 - y_1) + y_2 = 0$$

$$30y_3 - 49y_2 + 20y_1 = 0$$

$i=3$

$$25[y_4 - 2y_3 + y_2] + 5[y_4 - y_2] + y_3 = 0$$

$$30y_4 - 49y_3 + 20y_2 = 0$$

$i=4$

$$25[y_5 - 2y_4 + y_3] + 5[y_5 - y_3] + y_4 = 0$$

$$30y_5 - 49y_4 + 28y_3 = 0$$

As $y_0 = 0$ & $y_5 = 1$

S₀.

$$-49y_1 + 30y_2 = 0$$

$$20y_1 - 49y_2 + 30y_3 = 0$$

$$20y_2 - 49y_3 + 30y_4 = 0$$

$$20y_3 - 49y_4 = -30$$

$$y_1 = 0.44925$$

$$y_2 = 0.733776$$

$$y_3 = 0.899001$$

$$y_4 = 0.979184$$